Digital Chip Design Program

RTL Design

Task #06

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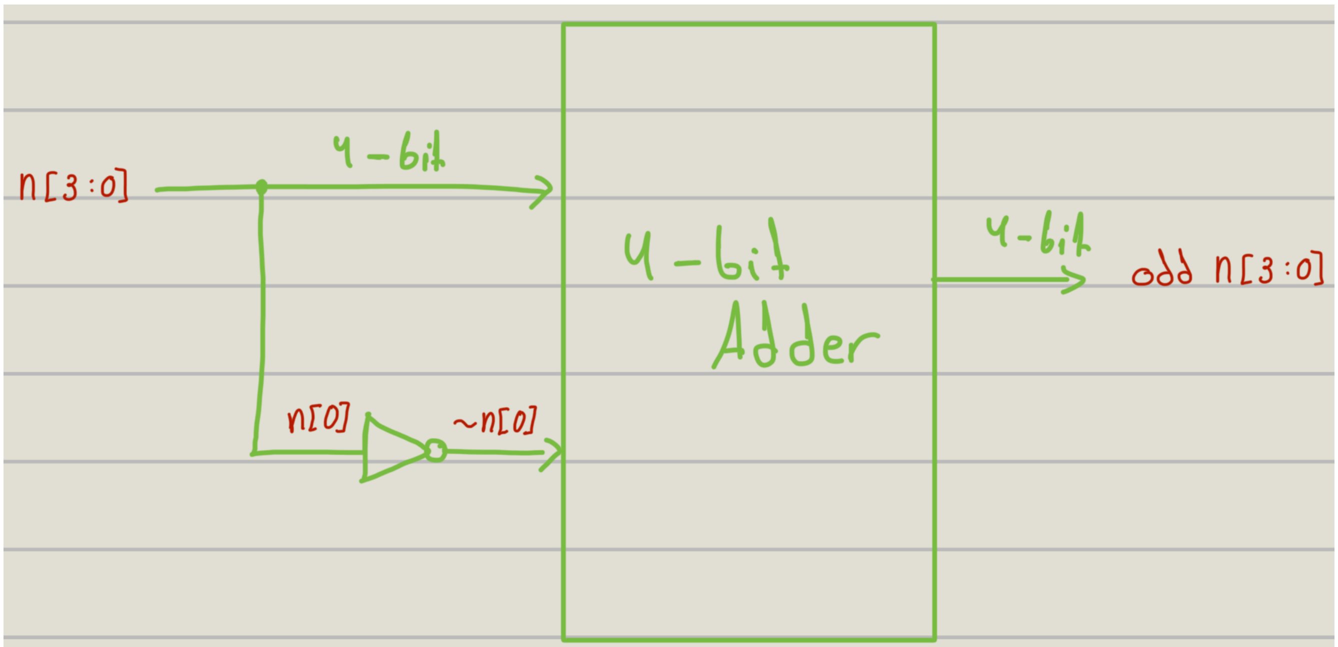
Combinational Datapath Design Part 2

Version 1.0

Task #1.1

1.1.1.2

Microarchitecture

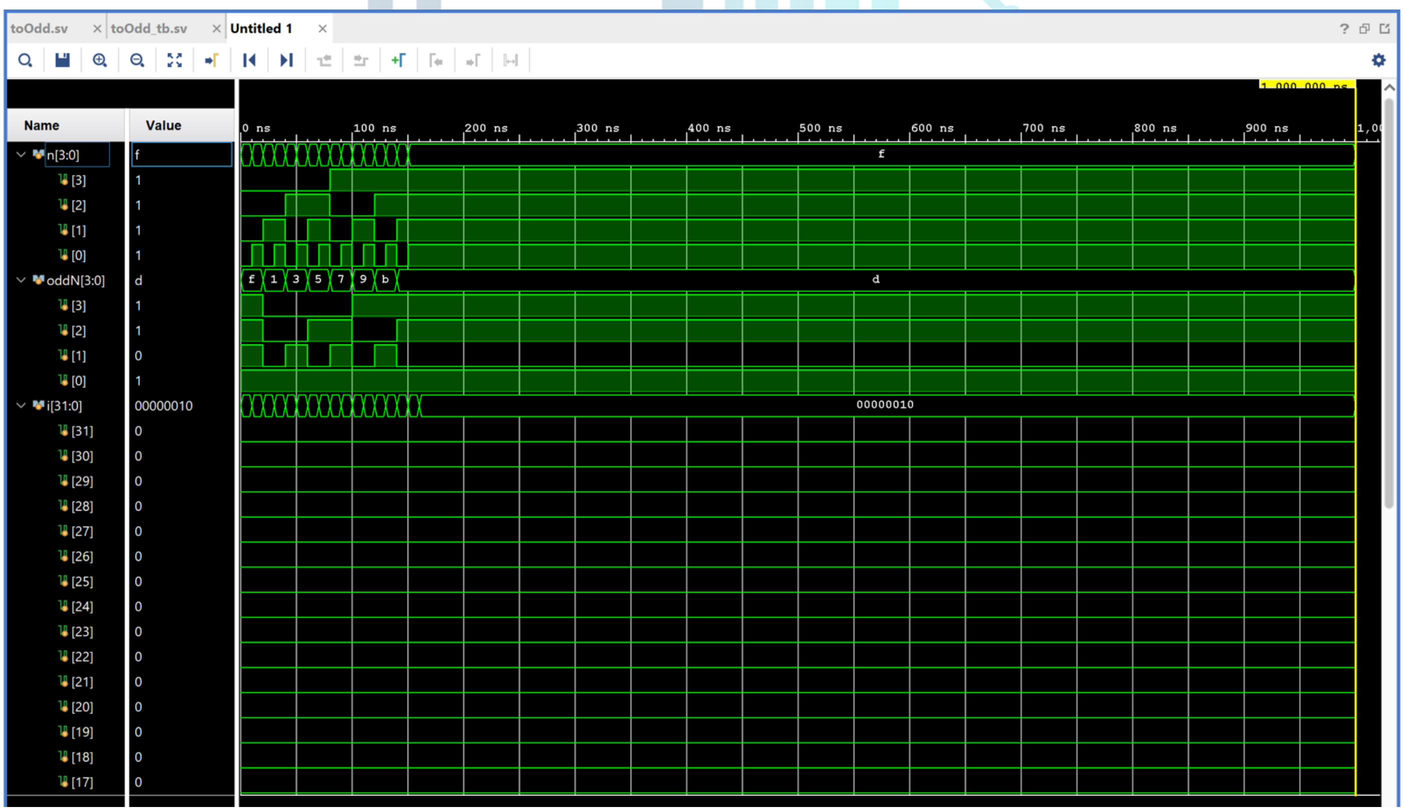


1.1.1.3

```
module toOdd (
    input  logic [3:0] n,
    output logic [3:0] oddN
);

    assign oddN = n + (~n[0]);

endmodule
```

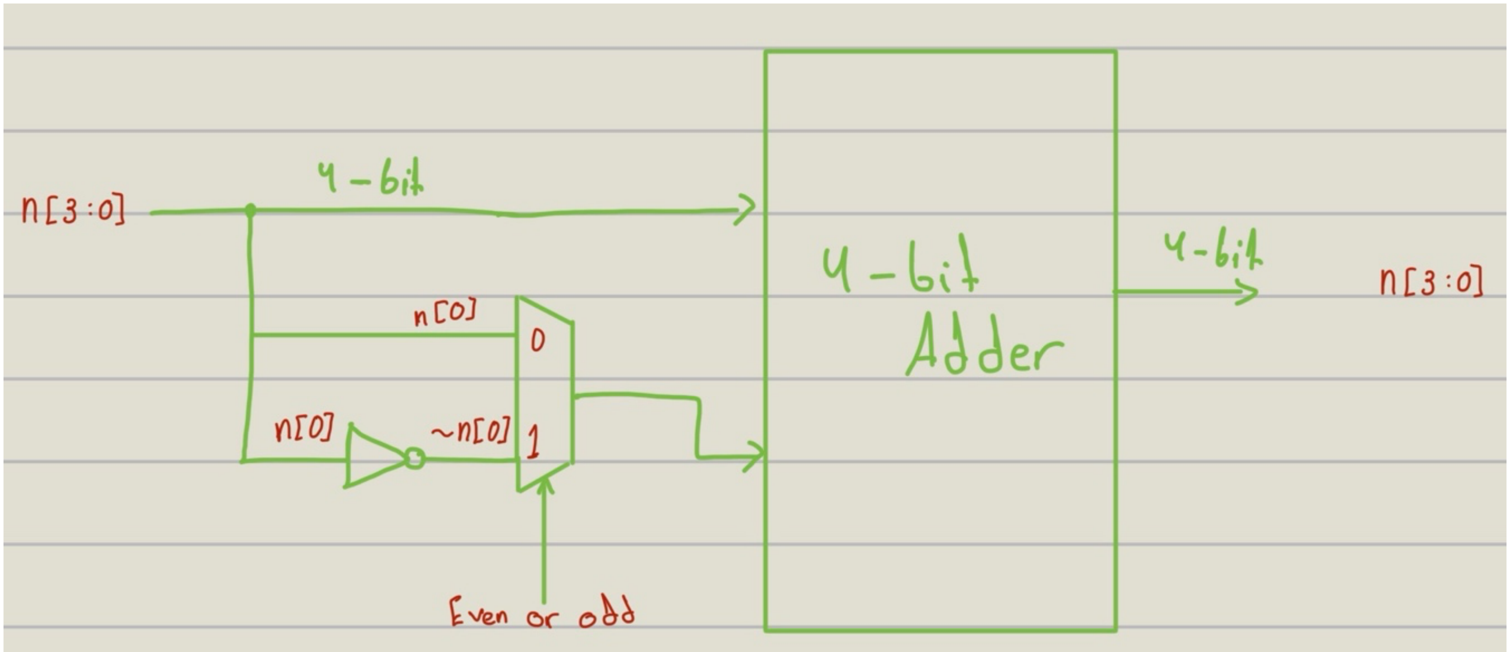
1.1.1.4

```
`timescale 1ns / 1ps
module toOdd_tb();
    logic [3:0] n;
    logic [3:0] oddN;

    toOdd dut (
        .n(n),
        .oddN(oddN)
    );

    int i;
    initial begin
        for (i = 0; i < 16; i = i + 1) begin
            n = i;
            #10;
        end
      #10  $finish;
    end
endmodule
```



1.1.2.2

Microarchitecture

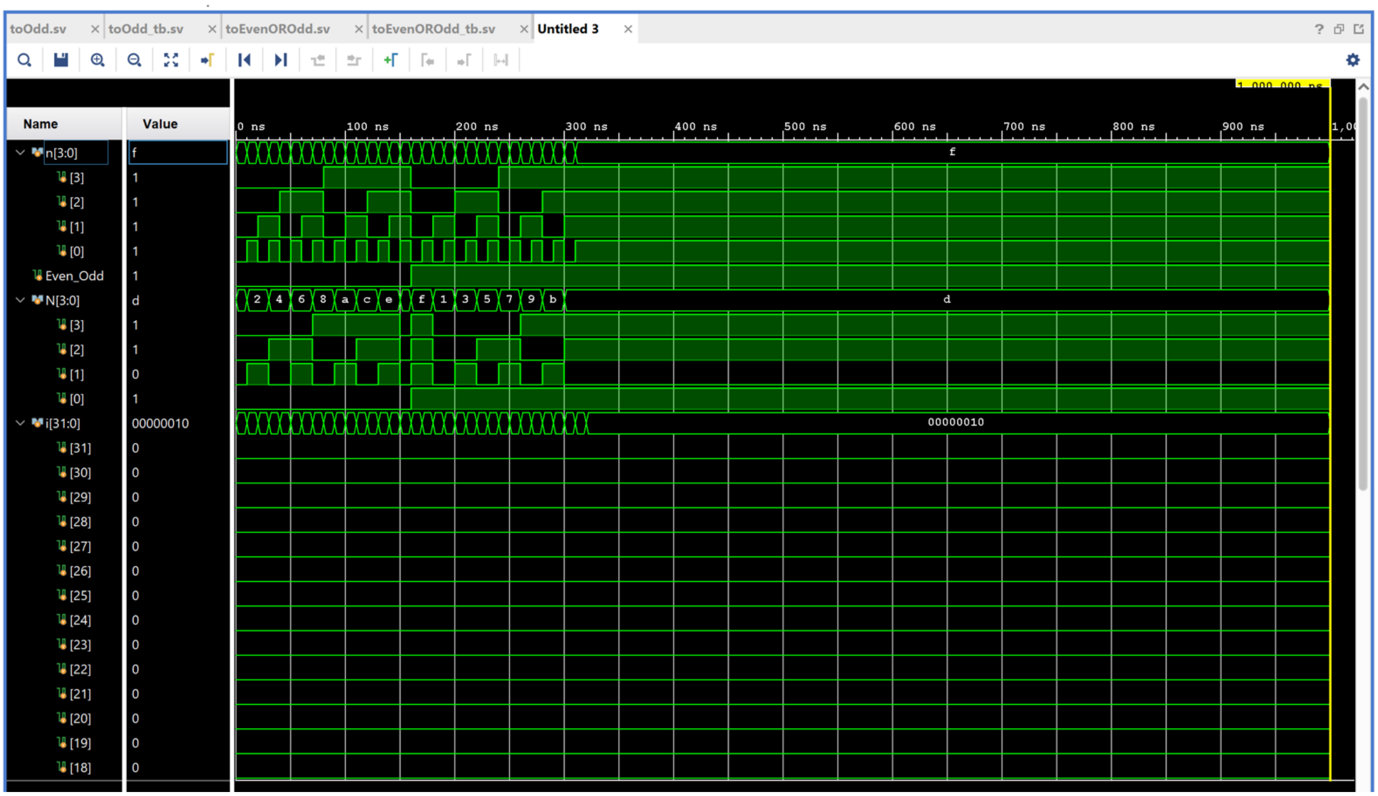


1.1.2.3

```
module toEvenOROdd (
    input  logic [3:0] n,
    input  logic       Even_Odd,
    output logic [3:0] N
);

    assign N = n + (Even_Odd ? (~n[0]) : n[0]);

endmodule
```

1.1.2.4

```
`timescale 1ns / 1ps

module toEvenOROdd_tb;
    logic [3:0] n;
    logic       Even_Odd;
    logic [3:0] N;

    toEvenOROdd dut (
        .n(n),
        .Even_Odd(Even_Odd),
        .N(N)
    );

    int i;
    initial begin
        Even_Odd = 0;
        for (i = 0; i < 16; i = i + 1) begin
            n = i;
            #10;
        end

        Even_Odd = 1;
        for (i = 0; i < 16; i = i + 1) begin
            n = i;
            #10;
        end

      #10 $finish;
    end
endmodule
```



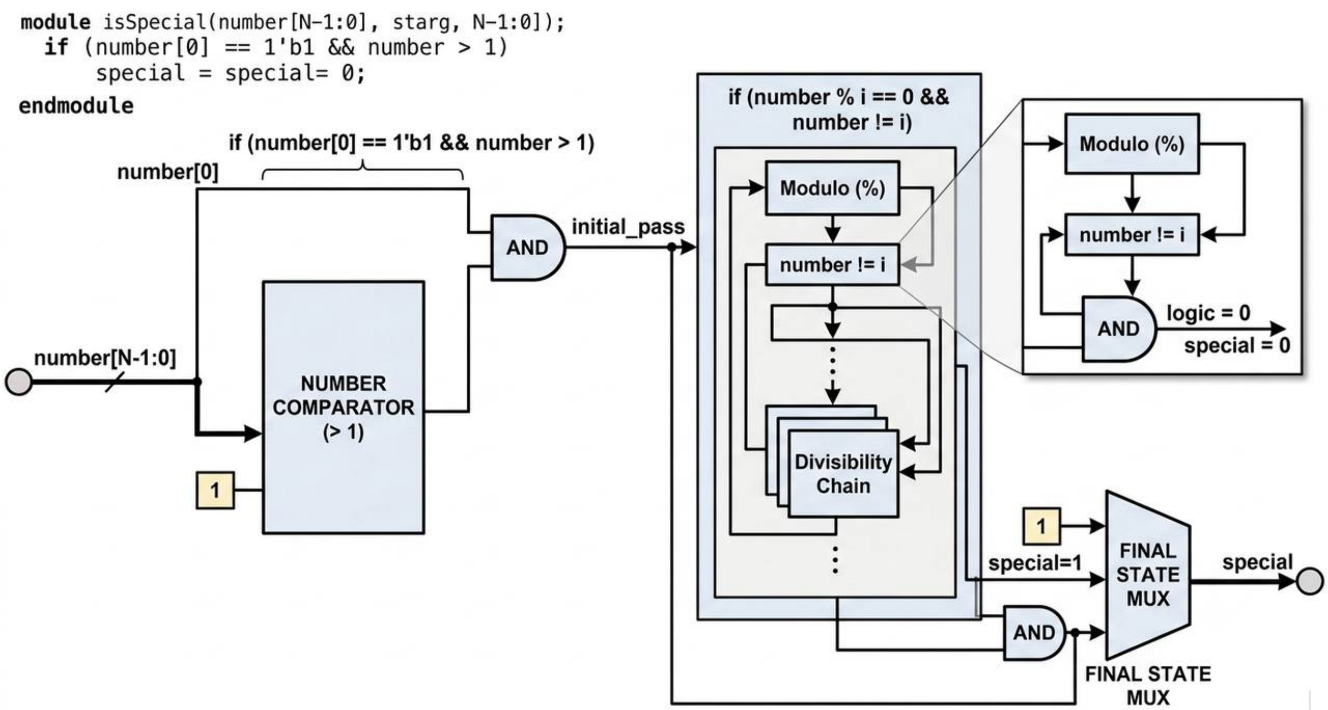
Task #1.2

1.2.1.1

```
    input  logic [N-1:0] number,
    output logic    special);
```

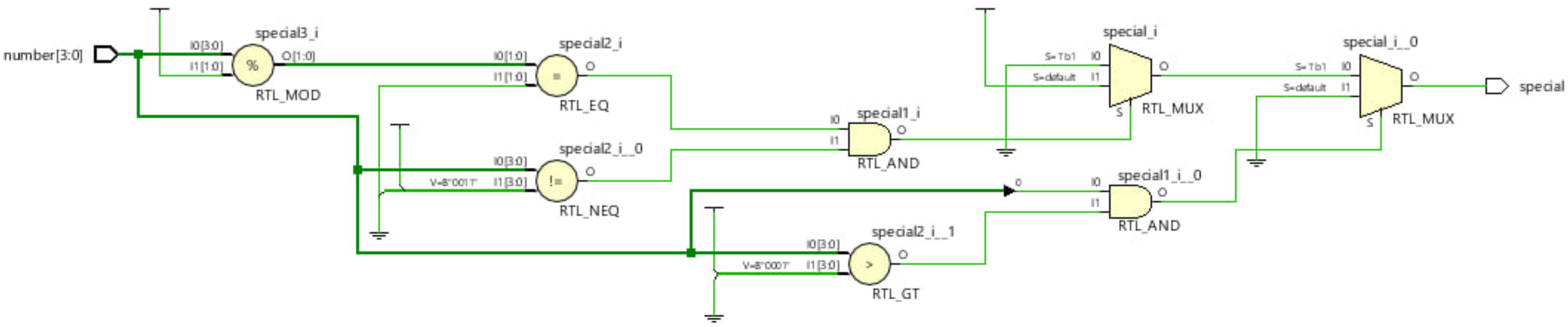
1.2.1.2

Microarchitecture



1.2.1.3

```
module isSpecial #(parameter N = 4)(
    input  logic [N-1:0] number,
    output logic    special);

    always_comb begin
        special = 1'b0;

        if (number[0] == 1'b1 && number > 1) begin
            special = 1'b1;
            for (int i = 3; i * i <= (1 << N); i += 2) begin
                if (number % i == 0 && number != i) begin
                    special = 1'b0;
                end
            end
        end
    end

endmodule
```

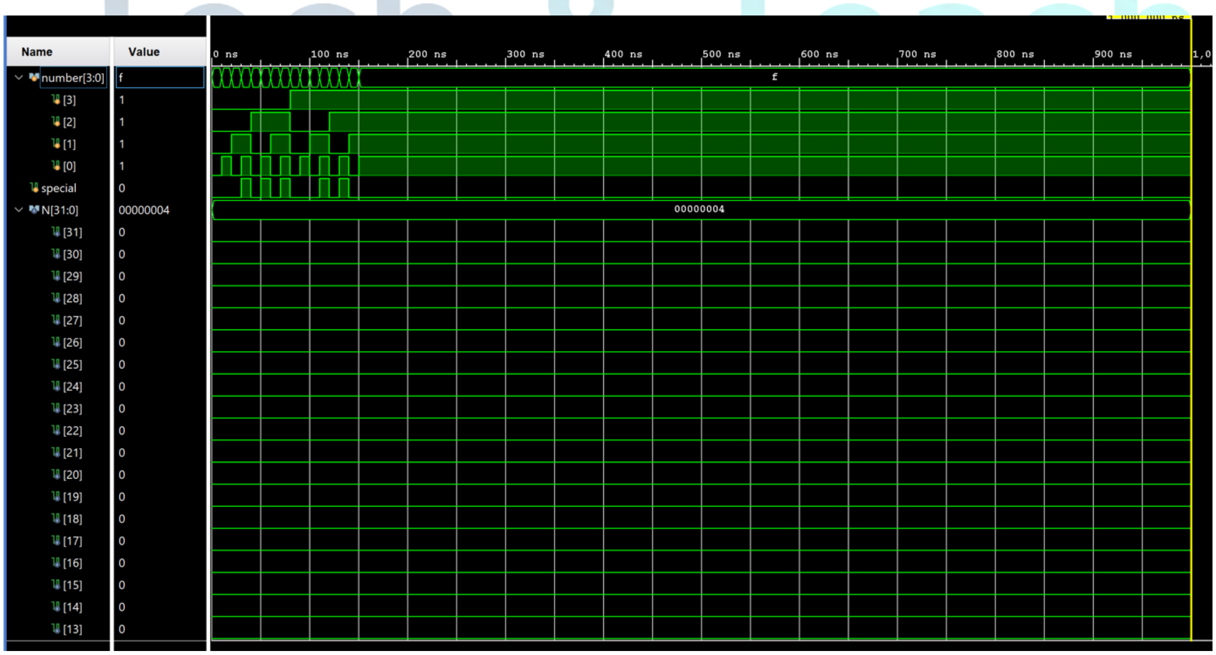


1.2.1.4

Project Summary | isspecial_tb.sv | Schematic

C:/Users/farah/Task2Day7/Task2Day7.srcs/sim_1/new/isspecial_tb.sv

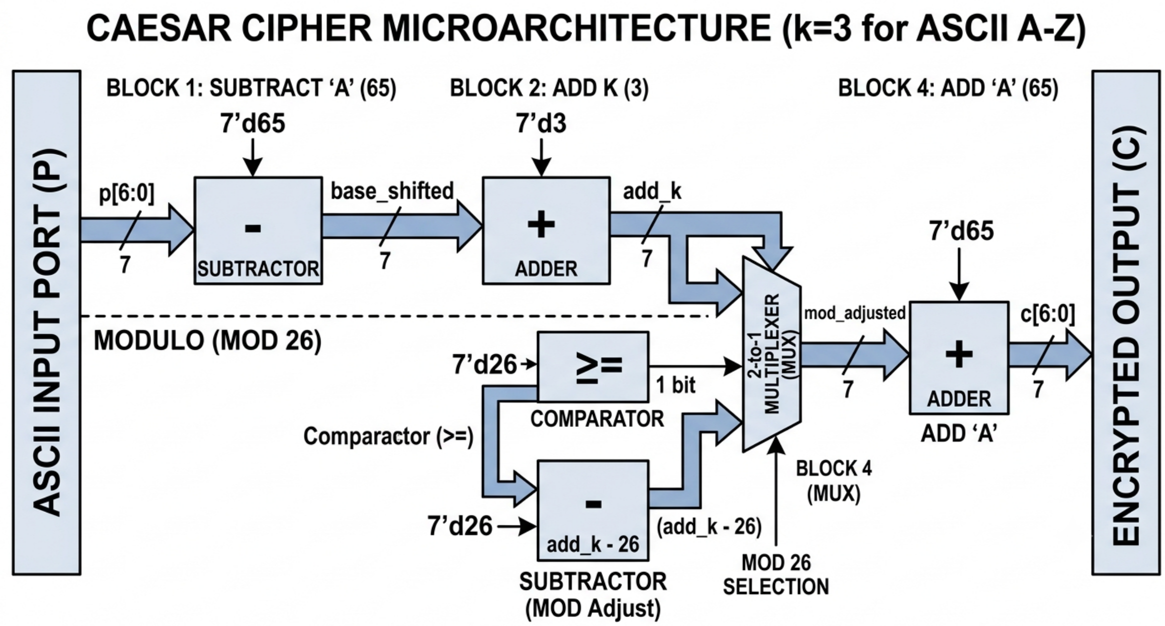
```
`timescale 1ns / 1ps
module isspecial_tb;
    parameter N = 4;
    logic [N-1:0] number;
    logic         special;

    isSpecial #(.N(N)) uut (
        .number(number),
        .special(special)
    );

    initial begin
        for (int i = 0; i < (1 << N); i++) begin
            number = i;
            #10;
        end
        $finish;
    end
endmodule
```



Task #1.3

1.3.1.2

Microarchitecture

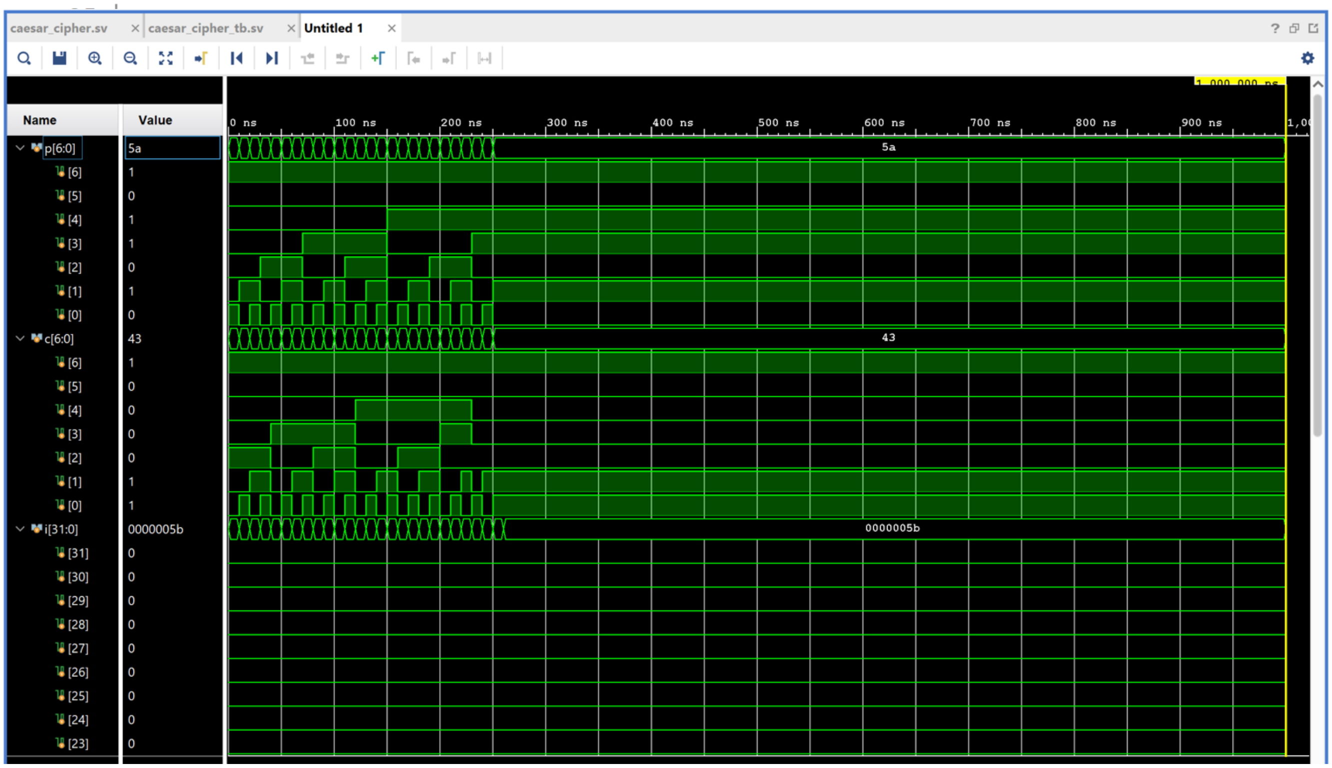


1.3.1.3

```
module caesar_cipher (
    input  logic [6:0] p,
    output logic [6:0] c
);

    logic [6:0] base_shifted;
    logic [6:0] add_k;
    logic [6:0] mod_adjusted;

    always_comb begin
        base_shifted = p - 7'd65;
        add_k = base_shifted + 7'd3;

        if (add_k >= 7'd26) begin
            mod_adjusted = add_k - 7'd26;
        end else begin
            mod_adjusted = add_k;
        end

        c = mod_adjusted + 7'd65;
    end

endmodule
```

1.3.1.4

```
`timescale 1ns / 1ps
module caesar_cipher_tb();

    logic [6:0] p;
    logic [6:0] c;

    caesar_cipher dut (
        .p(p),
        .c(c)
    );

    int i;

    initial begin
        for (i = 65; i <= 90; i = i + 1) begin
            p = i;
            #10;
        end

        #50;
       #10 $finish;
    end

endmodule
```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	